

Day 4 started by covering modules in python, specifically the pseudo random number generator module. This was used to create a simple “heads or tails” coin flip as seen in Figures 1 and 2.

Figure 1: Code utilizing random module

```
import random
rand_num = random.randint(a: 1, b: 10)
print(rand_num)

rand_num_0_to_1 = random.random()
print(rand_num_0_to_1)

random_float = random.uniform(a: 1, b: 10)
print(random_float)

heads_or_tails = random.randint(a: 1, b: 2)

if heads_or_tails == 1:
    print("Heads")
else:
    print("Tails")
```

Figure 2: Results of running the code in Figure 1

```
3
0.4571710670563053
8.545759121613894
Heads

Process finished with exit code 0
```

The next part of the lesson covered the Data Type lists. Figure 3 displays the various ways lists were used with Figure 4 showing the results of the code.

Figure 3

```
fruits = ["Apple", "Orange", "Banana"]
print(fruits[1])

# Negative Index
print(fruits[-1])

# Modifying List
fruits[2] = "Cherry"
print(fruits[2])
💡
#Appending List
fruits.append("Pear")
print(fruits)
```

Figure 4

```
Orange
Banana
Cherry
['Apple', 'Orange', 'Cherry', 'Pear']

Process finished with exit code 0
```

The concepts from random module and lists were put together to decide a random name from a list of people to pay the bill.

Figure 5

```
import random
friends = ["Alice", "Bob", "Charlie", "David", "Emanuel"]

random_friend = random.choice(friends)
print(random_friend)
```

The final project for Day 4 was creating rock, paper, scissors. This required utilizing lists, random module, conditionals, and game logic. The code can be found in the [main.py](#) file for this Day. The results of code can be seen in Figure 6.

Figure 6

```
Welcome to Rock, Paper, Scissors!
Type 0 for Rock, 1 for Paper, 2 for Scissors: 2
You chose
      -----
---'  _____)
           _____)
           _____)
           (_____)
---.__(_____)

Computer chose
      -----
---'  _____)
           (_____)
           (_____)
           (_____)
---.__(_____)

You lose!
Welcome to Rock, Paper, Scissors!
Type 0 for Rock, 1 for Paper, 2 for Scissors:
```